$$y'' - 2y' = e^{2t}\cos(t) \; ; \; y(0) = 0 \; ; \; y'(0) = 0$$

$$\mathcal{L}\{y''\} - 2\mathcal{L}\{y'\} = \mathcal{L}\{e^{2t}\cos(t)\}$$

$$\mathcal{L}\{y''\} = s^2 Y(s) - sy(0) - y'(0)$$

$$\mathcal{L}\{y'\} = sY(s) - y(0)$$

$a = 2$ y $b = 1$

$$\mathcal{L}\{e^{at}\cos(bt)\} = \frac{s-a}{(s-a)^2+b^2} \rightarrow \mathcal{L}\{e^{2t}\cos(t)\} = \frac{s-2}{(s-2)^2+1}$$

$$[s^2 Y(s) - sy(0) - y'(0)] - 2[sY(s) - y(0)] = \frac{s-2}{(s-2)^2+1}$$

$$s^2 Y(s) - 2sY(s) = \frac{s-2}{(s-2)^2+1}$$

$$(s^2 - 2s)Y(s) = \frac{s-2}{(s-2)^2+1}$$

$$s(s-2)Y(s) = \frac{s-2}{(s-2)^2+1}$$

$$Y(s) = \frac{s-2}{s(s-2)[(s-2)^2+1]}$$

$$Y(s) = \frac{1}{s[(s-2)^2+1]}$$

$$Y(s) = \frac{1}{s(s^2-4s+5)}$$

$$\frac{1}{s(s^2-4s+5)} = \frac{A}{s} + \frac{Bs+C}{s^2-4s+5}$$

$$1 = A(s^2-4s+5) + (Bs+C)s$$
$$1 = As^2 - 4As + 5A + Bs^2 + Cs$$
$$1 = (A+B)s^2 + (-4A+C)s + 5A$$

$$5A = 1 \rightarrow A = \frac{1}{5}$$

$$A+B = 0 \rightarrow \frac{1}{5} + B = 0 \rightarrow B = \frac{-1}{5}$$

$$-4A + C = 0 \rightarrow -4\left(\frac{1}{5}\right) + C = 0 \rightarrow C = \frac{4}{5}$$

$$Y(s) = \frac{\frac{1}{5}}{s} + \frac{-\frac{1}{5} + \frac{4}{5}}{s^2-4s+5}$$

$$Y(s) = \frac{1}{5}\left(\frac{1}{s} - \frac{s-4}{s^2-4s+5}\right)$$

$$\frac{s-4}{(s-2)^2+1} = \frac{(s-2)-2}{(s-2)^2+1} = \frac{s-2}{(s-2)^2+1} - \frac{2}{(s-2)^2+1}$$

$$Y(s) = \frac{1}{5}\left[\frac{1}{s} - \left(\frac{s-2}{(s-2)^2+1} - \frac{2}{(s-2)^2+1}\right)\right]$$

$$Y(s) = \frac{1}{5}\cdot\frac{1}{s} - \frac{1}{5}\frac{s-2}{(s-2)^2+1} - \frac{2}{5}\frac{1}{(s-2)^2+1}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} = 1$$

$$\mathcal{L}^{-1}\left\{\frac{s-2}{(s-2)^2+1}\right\} = e^{2t}\cos(t)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+1}\right\} = e^{2t}\sin(t)$$

$$y(t) = \frac{1}{5} - \frac{1}{5}e^{2t}\cos(t) + \frac{2}{5}e^{2t}\sin(t)$$